

GATE SOLUTION IN MINING ENGINEERING

(Containing Solved Questions 2015 - 2007)

By

RUPESH KUMAR SAHU

LOVELY PRAKASHAN, DHANBAD

Gate Solution-2008

Q 1. The trace of the matrix is

$$\begin{bmatrix} 2 & 2 & 3 \\ 3 & 2 & 3 \\ 4 & 1 & 2 \end{bmatrix}$$

- (A) 6 (B) 7 (C) 8 (D) 9

Answer (A)

Solution .

Note- Trace of the matrix - sum of diagonal elements of a square matrix is called trace of matrix.

$$\therefore \text{Trace of the matrix} = 2 + 2 + 2 = 6$$

Q 2. If X is a continuous random variable and $f(x)$ defines its probability density function , then the expected value of X is

- (A) $\int_{-\infty}^{+\infty} f(x) dx$ (B) $\sum_{i=-\infty}^{+\infty} x_i$
(C) $\sum_{i=-\infty}^{+\infty} x_i f(x)$ (D) $\int_{-\infty}^{+\infty} x f(x) dx$

Answer (D)

Q 3. The tool used to correct borehole deviation is

- (A) string shot (B) Kelly (C) Whipstock (D) Rachet

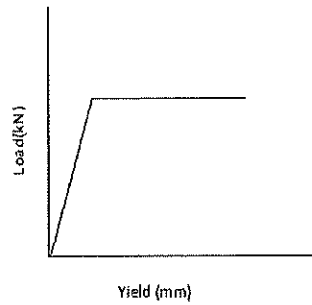
Answer (C)

Q 4. A phreatic surface experiences a pressure

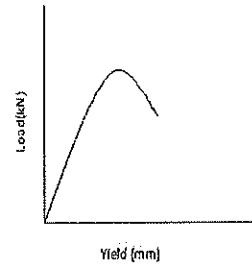
- (A) less than atmospheric pressure (B) equal to atmospheric pressure
(C) more than barometric pressure (D) less than barometric pressure

Answer (B)

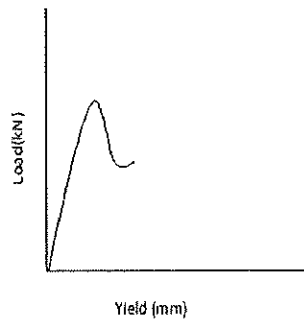
Q 5. The load yield characteristics of a hydraulic prop is represented by the curve



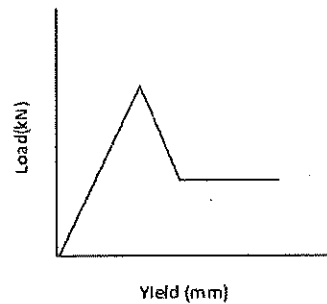
(A)



(B)



(C)



(D)

Answer (A)

Q 6. In longwall caving, the thickness of immediate roof is calculated from

- (A) Bulking factor and width of the longwall face**
- (B) Seam thickness and width of the longwall face**
- (C) Seam thickness and bulking factor**
- (D) Seam thickness and length of the panel**

Answer (C)

Q 7. During over-winding, a cage is safely suspended in the headgear due to

- (A) Bulk chain (B) Rope capel
(C) D-link (D) Detaching hook

Answer (D)

Q 8. Depending upon the decreasing ability of surrounding rock to store strain energy, the underground metal mining method can be ordered as

- (A) Cut and fill stoping, sublevel caving, sublevel open stoping , block caving
(B) Sublevel open stoping, cut and fill stoping, sublevel caving, block caving
(C) Sublevel caving, sublevel open stoping, cut and fill stoping, block caving
(D) Block caving, sublevel caving, sublevel open stoping, cut and fill stoping

Answer (B)

Q 9. If the swell factor of ore in shrinkage stoping is 1.4. the output from the stope in percentage of broken ore is

- (A) 0 (B) 29 (C) 40 (D) 100

Answer (B)

Solution .

$$\begin{aligned}\% \text{ of broken ore} &= \frac{\text{extra volume}}{\text{broken ore volume}} \times 100 \\ &= \frac{(1.4 - 1)}{1.4} \times 100 \\ &= 29 \%\end{aligned}$$

Q 10. The velocity of the wave type that determines the 'rippability' of rockmass is

- (A) P-wave (B) S -wave (C) Raleigh wave (D) Love wave

Answer (A)

Q 11. In the order of the chronological development, the longwall support system are arranged as

P. Powered support

Q. Link bar and friction support

R. Frame support

S. Hydraulic support

(A) $P > Q > R > S$ (B) $R > S > Q > P$ (C) $S > R > P > Q$ (D) $Q > S > R > P$

Answer (B)

Q 12. Effective temperature is estimated from

(A) Wet bulb temperature, relative humidity and air velocity

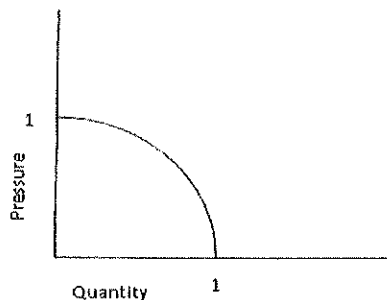
(B) Dry bulb temperature, relative humidity and air velocity

(C) Dry bulb temperature, wet bulb temperature and air velocity

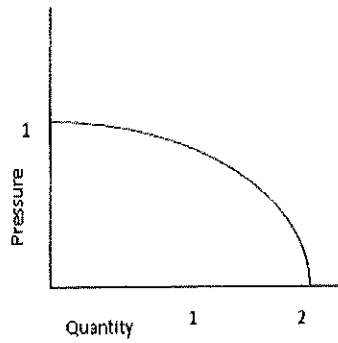
(D) Dry bulb temperature, wet bulb temperature and relative humidity

Answer (C)

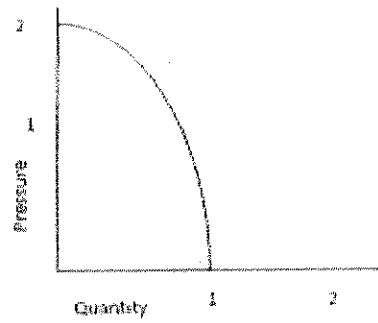
Q 13. Pressure –quantity characteristics of a mine fan is given below:



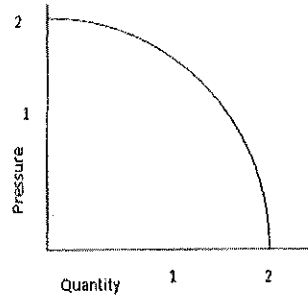
The combined characteristics of two such identical fan installed in parallel is



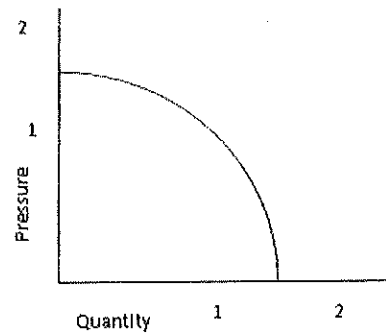
(A)



(B)



(C)



(D)

Answer (A)

Q 14 Under identical water head and roadways condition for water dam construction, if P,Q and R represent the thickness of flat dem; cylindrical dam and spherical dam respectively. The thickness would follow the order

- (A) $R > P > Q$ (B) $P > R > Q$ (C) $P > Q > R$ (D) $Q > P > R$

Answer (C)

Q 15. The grain size distribution of soil is known as

- (A) Permeability (B) Structure (C) Porosity (D) Texture

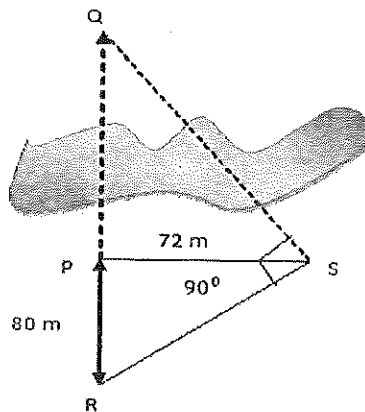
Solution (B)

Q 16. Electrostatic precipitator works on the principle of

- (A) Capacitance change (B) Ionization of particles
(C) Electro heating of gases (D) Centrifuging the gaseous molecules

Answer (B)

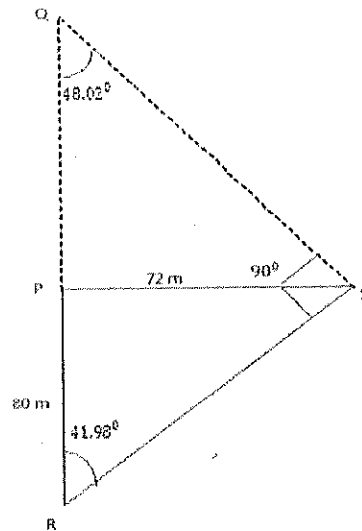
Q 17. In the figure shown below , the distance RP and PS are measured to be 80 m and 72 m respectively. The distance PQ in m



- (A) 60.4 (B) 66.4 (C) 64.8 (D) 68.4

Answer (C)

Solution .



$$\tan \angle PRS = \frac{72}{80}$$

$$\therefore \angle PRS = 41.98^\circ$$

$$\therefore \angle PQR = 180^\circ - 90^\circ - 41.98^\circ$$

$$\therefore \angle PQR = 48.02^\circ$$

Now, In $\triangle PSQ$

$$\tan 48.02^\circ = \frac{72}{PQ}$$

$$\therefore PQ = 64.8 \text{ m}$$

Q 18. In PERT network, the activity duration is assumed to follow

- | | |
|-------------------------|---------------------------|
| (A) Beta distribution | (B) Binomial distribution |
| (C) Normal distribution | (D) Weibull distribution |

Answer (A)

Q 19. For an LP problem, identify the INCORRECT statement

- (A) Optimal points lies in one of the corner points
- (B) Objective function is linear

(C) All the constraints are linear

(D) Optimal point lies in any of the interior points of the feasible region

Answer (D)

Q 20. In an bi-axial stress field the vertical stress is 10 MPa and the poisson ratio for the rock mass is 0.2. The horizontal stress in MPa is,

(A) 1.5 (B) 2.5 (C) 2.0 (D) 5.0

Answer (B)

Solution .

We have,

$$\frac{\text{horizontal stress}}{\text{vertical stress}} = \frac{\text{poisson ratio}}{1 - \text{poisson ratio}}$$

$$\therefore \frac{\text{horizontal stress}}{10} = \frac{0.2}{1 - 0.2}$$

$$\therefore \text{horizontal stress} = 2.5 \text{ MPa}$$

Q 21. Given :-

Bench height - 12 m

Burden - 4 m

Spacing - 5 m

Sub- grade drilling - 2 m

Explosive per hole - 120 kg

Density of rock - 2600 kg/m^3

The powder factor in tonne/kg is

(A) 2.0 (B) 4.6 (C) 5.2 (D) 7.3

Answer (C)

Solution .

We have,

$$\begin{aligned}
 \text{Powder Factor} &= \frac{\text{blasted material (tonne)}}{\text{explosive required (kg)}} \\
 &= \frac{\text{hole length} \times \text{spacing} \times \text{burden} \times \text{rock density (tonne)}}{\text{explosive required (kg)}} \\
 &= \frac{12 \times 4 \times 5 \times 2600/1000}{120} \\
 &= 5.2
 \end{aligned}$$

Q 22. Match the following

Equipment	slice thickness (range in m)	Action
P. Dragline	1. 6 - 12	a. Crowding
Q. Shovel	2. 30 - 40	b. Hoisting
R. Surface miner	3. 0.2 - 0.4	c. Cutting
(A) P-1-b; Q-2-a; R-3-c		(B) P-2-b; Q-1-a; R-3-c
(C) P-2-a; Q-1-b; R-3-c		(D) P-1-b; R-1-a; Q-3-c

Answer (B)

Q 23. Given:

Value of ore : Rs.600 per tonne

Production cost : Rs.600 per tonne

Cost of overburden removal : Rs. 50 per m^3

The break- even stripping ratio in m^3 /tonne is

(A) 4:1 (B) 3:1 (C) 1:3 (D) 1:4

Answer (A)

Solution .

We know,

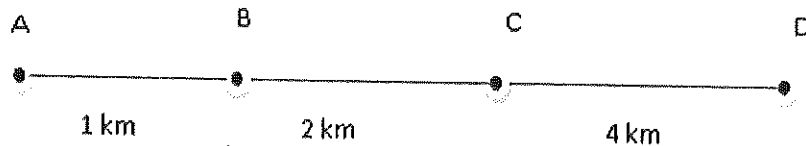
Break even stripping ratio

$$= \frac{\text{recoverable price per tonne of ore} - \text{O.C production cost per tonne of ore}}{\text{cost of extraction per cubic meter of overburden material}}$$

$$= \frac{600 - 400}{50}$$

$$= 4:1$$

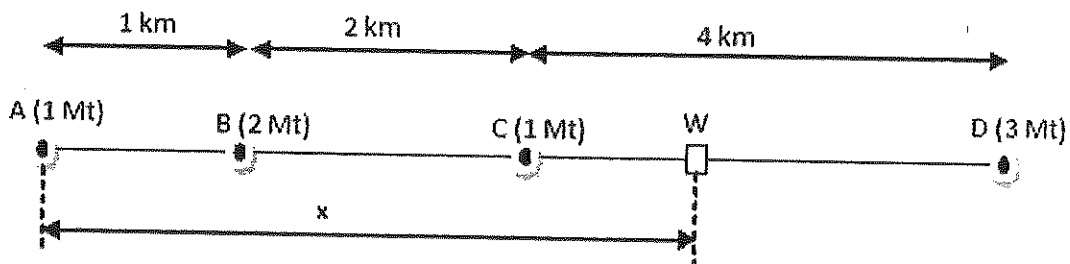
- Q 24. Four mines A, B, C and D are located along a road as shown with production in Mt per year 1, 2, 1 and 3 respectively. In order to handle total coal produced, the distance of the coal washery along the road from the mine A in km is



- (A) 4.01 (B) 3.91 (C) 3.81 (D) 3.71

Answer (D)

Solution .



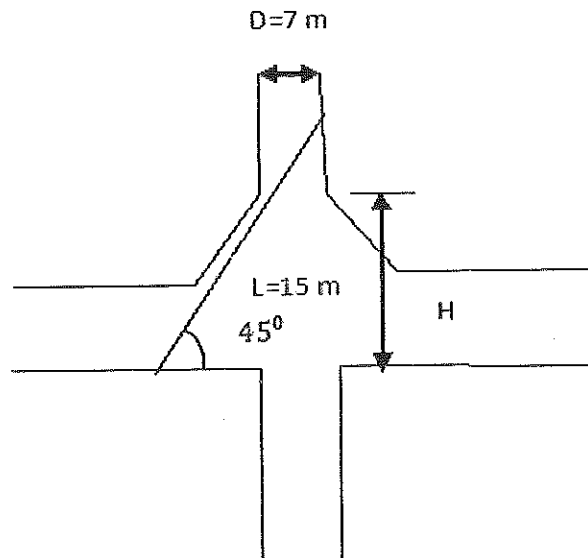
Where, x - ideal distance from coal washery from mine A .

$$\therefore x \times 1 + 2 \times (x - 1) + 1 \times (x - 3) = 3 \times (7 - x)$$

$$x + 2x - 2 + x - 3 = 21 - x$$

$$\therefore x = 3.71 \text{ km}$$

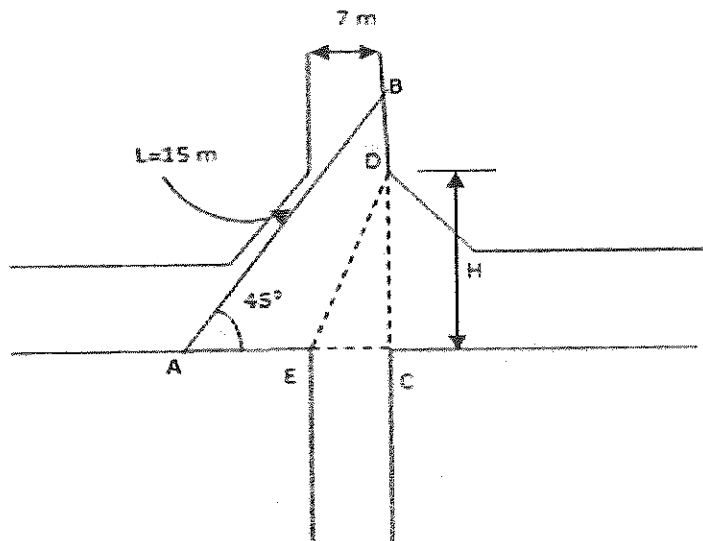
Q 25. A shaft inset is as shown below. To transport a 15 m long object, the height 'H' of the inset in m should be



- (A) 10.5 (B) 7.0 (C) 6.5 (D) 5.9

Answer (B)

Solution .



In $\triangle ABC$

$$\cos 45^\circ = \frac{AC}{15}$$

$$\therefore AC = 10.606 \text{ m}$$

$$\text{And } BC = \sqrt{(15)^2 - (10.606)^2} = 10.606 \text{ m}$$

In $\triangle ABC$ & $\triangle CDE$

$$\therefore \frac{H}{EC} = \frac{BC}{AC}$$

$$\frac{H}{7} = \frac{10.606}{10.606}$$

$$\therefore H = 7 \text{ m}$$

Q 26. Match the following

Blast problem

Cause

P. Misfire

1. Poor stemming

Q. Vibration

2. Low current

R. Blown-out-shot

3. Excess charge

S. Cut-off shot

4. Improper delays

(A) P-3, Q-2, R-4, S-1

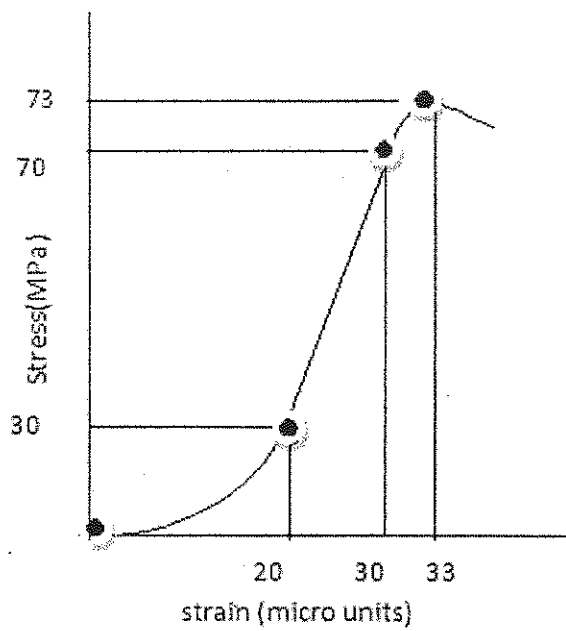
(B) P-4, Q-1, R-2, S-3

(C) P-2, Q-3, R-1, S-4

(D) P-1, Q-2, R-4, S-3

Answer (C)

Q 27. From the stress-strain diagram shown below, the tangent and the secant moduli of elasticity in GPa are



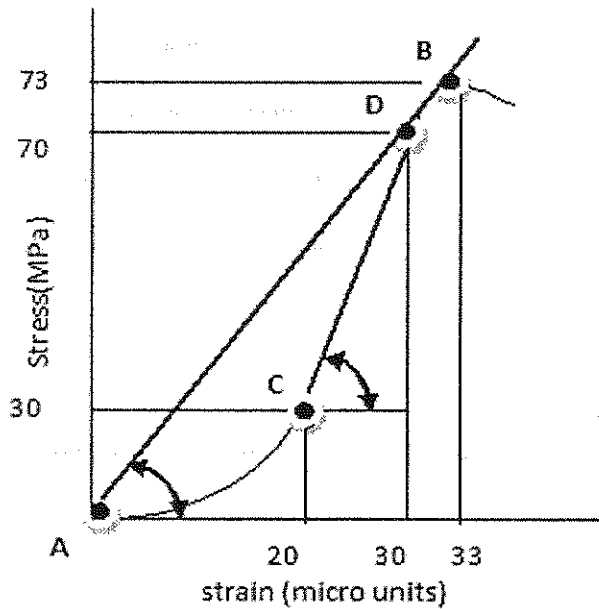
(A) 4.0, 2.2

(B) 3.3, 2.3

(C) 3.3, 1.5

(D) 4.0, 1.5

Answer (A)



Tangent - tangent to a plane curve at a given point is the straight line that touches the curve

at that point.

Secant- secant to a curve is a line that intersect two point on the curve.

∴ In above figure CD is tangent and AB is secant

Hence,

$$\text{Tangent modulus} = \frac{70 - 30}{30 - 20} = \frac{40}{10} = 4$$

$$\text{Secant modulus} = \frac{73}{33} = 2.2$$

- Q 28.** A bord and pillar operatios ia planned at a depth of 300 m in a strata of average unit weight 24.5 kN/m^3 and compressive strength 15.50 MPa . If the width of opening is 6 m considering a factor of safety of 1 , the maximum possible extraction ratio in percentage is
- (A) 28 (B) 34 (C) 45 (D) 53

Answer (D)

Solution .

$$\text{Strength of pillar} = 15.5 \text{ MPa} = 15.5 \times 10^6 \text{ N/m}^2$$

$$\text{Depth} = 300 \text{ m}$$

$$\text{Unit weight (w)} = 24.5 \text{ kN/m}^3 = 24.5 \times 10^3 \text{ N/m}^3$$

$$\text{Width of gallery or opening (b)} = 6 \text{ m}$$

$$\text{Factor of safety} = \frac{\text{strength of pillar}}{\text{load acting on pillar}}$$

$$\text{And in case of square pillar ,load acting on pillar} = w.D \frac{(a+b)^2}{a^2}$$

$$\therefore 1 = \frac{15.5 \times 10^6}{w.D \frac{(a+b)^2}{a^2}}$$

$$\therefore 1 = \frac{15.5 \times 10^6}{24.5 \times 10^3 \times 300 \frac{(a+6)^2}{a^2}}$$

On solving, we found that

$$a = 13.268 \text{ m}$$

Now,

$$\begin{aligned} \text{Extraction ratio (\%)} &= 1 - \frac{a^2}{(a+b)^2} \times 100 \\ &= 1 - \frac{13.268^2}{(13.268+6)^2} \times 100 \\ &= 53 \% \end{aligned}$$

Q 29. Match the following

Stoping method	Ore handling system	Support system
P. Breast stoping	1. LHD	a. In situ pillar
Q. Cut and fill	2. Scraper	b. Unsupported
R. Sublevel stoping	3. Gravity flow	c. Mill tailings
(A) P-2-a; Q-1-c; R-3-b	(B) P-1-a; Q-3-c; R-2-b	
(C) P-2-b; Q-1-a; R-3-c	(D) P-1-c; Q-3-a; R-2-b	

Answer (A)

Q 30. Match the following

Access	Haulage	Mineralisation location
P. Shaft	1. Track	a. moderate depth
Q. Decline	2. Trackless	b. Deep seated
R. Adit	3. Hoisting	c. Hillock

- | | |
|-------------------------|-------------------------|
| (A) P-1-a; Q-3-b; R-2-c | (B) P-3-b; Q-2-a; R-1-c |
| (C) P-2-a; Q-1-b; R-3-c | (D) P-2-b; Q-3-c; R-1-a |

Answer (B)

Q 31. Match the following

Mining method	Operation
P. Bord and pillar	1. Longhole radial drilling
Q. Sublevel caving	2. Splitting and slicing
R. Longwall retreating	3. Loosening under strata pressure
S. Integrated Caving	4. Mechanical cutting
(A) P-1, Q-4, R-3, S-2	(B) P-2, Q-3, R-1, S-4
(C) P-4, Q-2, R-3, S-1	(D) P-2, Q-1, R-4, S-3

Answer (D)

Q 32. A 30 m tape has an error of ± 0.005 m. If a length of 1500 m is measured with this tape, the expected total error made in the measurement in m is

- (A) ± 0.025 (B) ± 0.030 (C) ± 0.035 (D) ± 0.04

Answer (C)

Solution.

$$\begin{aligned}\text{The expected total error made in the measurement is} &= \pm 0.005 \times \sqrt{\frac{1500}{30}} \\ &= \pm 0.035\end{aligned}$$

Q 33. Match the following

Instrument	Principal features	Application
P. Tilting level	1. Micrometer	a. Levelling
Q. Microoptic theodolite	2. Magnetic needle	b. Traversing
R. Telescopic alidade	3. U-tube	c. Azimuth
S. Compass	4. Plane table surveying	d. Contouring

(A) P-1-b; Q-2-c; R-4-a; S-3-d

(B) P-4-b; Q-3-a; R-1-c; S-2-d

(C) P-2-c; Q-3-b; R-4-a; S-1-d

(D) P-3-a; Q-1-b; R-4-d; S-2-c

Answer (D)

Q 34. A confined aquifer of 75 m thickness has 2 monitoring wells spaced 2500 m apart along the direction of water flow. The hydraulic conductivity of the aquifer is 40 m per day. The water head difference between the wells is 1.5 m. applying the Darcy law's, the rate of flow per minute of distance perpendicular to the direction of flow in m^3/day is

(A) 2.1

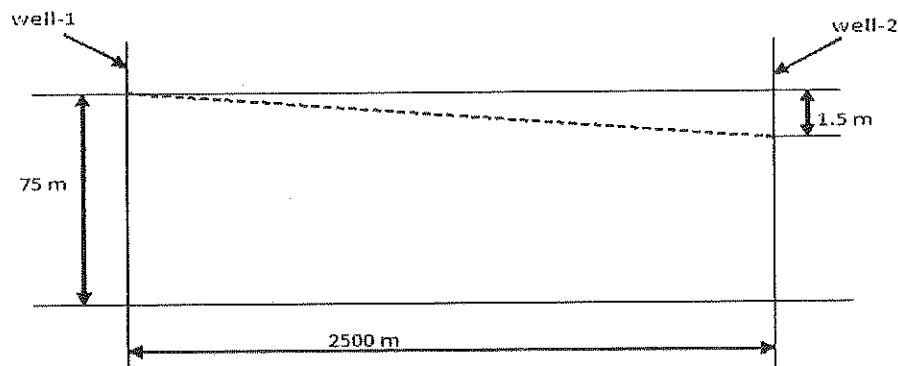
(B) 1.8

(C) 1.45

(D) 1.2

Answer (B)

Solution .



We have from Darcy law,

$$Q = kiA$$

Where,

k- hydraulic conductivity = 40 m/day

A - Cross sectional area = $75 \times 1 = 75 m^2$

$$i - \text{hydraulic gradient} = \frac{h}{L} = \frac{1.5}{2500}$$

[h- difference of hydraulic head over distance L]

Hence,

$$Q = 40 \times \frac{1.5}{2500} \times 75$$

$$Q = 1.8 \text{ m}^3/\text{day}$$

Q 35. Precipitation of the metallic ion in mine water drainage is carried out by

(A) CaSO_4 and MgSO_4 (B) CaCO_3 and MgCO_3

(C) Ca(OH)_2 and NaOH (D) CaCO_3 and MgSO_4

Answer (D)

Q 36. A drum winder of radius 2.5 m draws a power of 308 KW when the maximum rope speed is 7 m/s. The RMS torque in kNm is

(A) 55 (B) 76 (C) 110 (D) 144

Answer (C)

Solution .

$$\text{Power} = \text{force} \times \text{velocity}$$

$$308 \times 1000 = \text{force} \times 7$$

$$\therefore \text{force} = 44000 \text{ N}$$

$$\therefore \text{The RMS torque in kNm} = \text{force} \times \text{radius}$$

$$= 44000 \times 2.5$$

$$= 110 \text{ kNm.}$$

Q 37. A belt conveyor conveys material of average cross-sectional area of 0.09 m^2 of bulk density 1.5 toonne/m^3 , at a speed 2 m/s. The carrying capacity of the belt in tonne/hr is

(A) 972 (B) 864 (C) 732 (D) 643

Answer (A)

Solution .

Carrying capacity (T) of belt conveyor is given by:-

$$T = a.b.v \text{ tonne/s}$$

Where,

$$a = \text{cross sectional area} = 0.09 \text{ m}^2$$

$$b = \text{bulk density} = 1.5 \text{ tonne/m}^3$$

$$v = 2 \text{ m/s}$$

$$T = 0.09 \times 1.5 \times 2$$

$$T = 0.27 \text{ tonne/s}$$

$$T = 972 \text{ tonne/hour}$$

Q 38. The wt % of solids in a sand-water stowing pipe is 60. If the solids density 3000 kg/m^3 , the pulp density of the slurry in kg/m^3

- (A) 1380 (B) 1420 (C) 1560 (D) 1670

Answer (D)

Solution .

We have,

$$\text{Pulp density} = \frac{\text{Mass of pulp}}{\text{Volume of pulp}}$$

$$\text{Pulp density} = \frac{\text{Mass of pulp}}{\text{Volume of water} + \text{Volume of sand}}$$

$$\text{Or pulp density} = \frac{\text{mass of pulp}}{\frac{\text{mass of water}}{\text{density of water}} + \frac{\text{mass of sand}}{\text{density of sand}}}$$

$$\text{Mass of sand} = \text{mass of pulp} \times 60 \% = \text{mass of pulp} \times 0.6$$

$$\text{Mass of water} = \text{mass of pulp} \times (100-60)\% = \text{mass of pulp} \times 0.4$$

$$\therefore \text{Pulp density} = \frac{\text{mass of pulp}}{\text{mass of pulp} \left[\frac{0.4}{1000} + \frac{0.6}{3000} \right]}$$

$$\therefore \text{Pulp density} = 1666.7 \text{ kg/m}^3$$

Q 39. A mining project comprising of A,B and C activities is scheduled for 90 days at a cost of Rs 1200 million. The manager of project decides to reduce the time for completion of the project to 85 days. The decision was taken after 45 days.



Activity	A	B	C
Duration (days)	40	15	35
Crashing cost /day (million rupees)	15	25	20

The minimum project cost in million rupees after crashing by 5 days is-

- (A) 1100 (B) 1300 (C) 1475 (D) 1825

Answer (B)

Solution.

Here, crashing of project is done along C activity duration because having minimum crashing cost as compared to B activity duration.

$$\therefore \text{minimum cost of project after 5 days crashing is} = 1200 + 5 \times 20 = 1300 \text{ million rupees}$$

Q 40. the following information is provided for an ore deposit

Number of waste blocks = 10

Number of ore blocks = 5

Volume of each waste block, m^3 = 600

Total cost of waste handling per m^3 = Rs. 100

Tonnage of each ore block = 400
 Total cost of ore handling per ton = Rs. 150
 Sale price of ore per ton = Rs.500

The net cash flow of mining the deposit in lakhs of rupees, is

(A) 3.4 (B) 2.5 (C) 1.0 (D) 0.8

Answer (C)

Solution .

Sale price of ore = $5 \times 400 \times 500 = 1000000$ rupees

Total cost of waste handling = $10 \times 600 \times 100 = 600000$ rupees

Total cost of ore handling = $5 \times 150 \times 400 = 300000$ rupees

Hence, net cash flow of mining deposit = $1000000 - 600000 - 300000 = 100000$ rupees
 = 1 lakh rupees

Q 41. Determine the correctness or otherwise of the following Assertion [a] and the reason [r]

Assertion : While stonedust barrier may be effective against a coal dust explosion , the same is not true is case of firedamp explosion

Reason : In general, firedamp explosion are much more powerful than coal dust explosions.

- (A) Both [a] and [r] are false
- (B) [a] is true but [r] is false
- (C) Both [a] and [r] are true and [r] is the correct reason for [a]
- (D) Both [a] and [r] are true but [r] is not the correct reason for [a]

Answer (B)

Q 42. Match the following

Component of flame safety lamp	Purpose of component
P. Asbestos rings	1. Dissipation of heat of flue gases
Q. Wire gauges	2. Formation of air tight joints

R. Outer glass

3. Arrest of explosion inside the lamp

S. Combustion chimney

4. Separation of inlet air from flue gases

(A) P-2, Q-1, R-3, S-4

(B) P-4, Q-1, R-2, S-3

(C) P-2, Q-4, R-3, S-1

(D) P-1, Q-2, R-4, S-3

Answer (A)

Q 43. A road header district produces 20 mg/m^3 of airborne dust with the following size distribution

Size up to	cumulative wt %
$1 \mu\text{m}$	1
$5 \mu\text{m}$	5
$10 \mu\text{m}$	10
$20 \mu\text{m}$	20
$50 \mu\text{m}$	50
$> 50 \mu\text{m}$	100

The concentration of respirable fraction of dust in mg/m^3 is

(A) 0.2

(B) 2.0

(C) 10.0

(D) 1.0

Answer (B)

Solution.

Respirable size of dust $\leq 10 \mu\text{m}$

$$\begin{aligned}\text{Hence, concentration of respirable fraction of dust} &= 20 \times \frac{10}{100} \\ &= 2 \text{ mg/m}^3\end{aligned}$$

Q 44. For a person working in atmosphere containing 21 % O_2 , the exhaled air contains 4.5 % CO_2 and 16 % O_2 . The respiratory quotient of breathing is

- (A) 0.21 (B) 0.9 (C) 0.28 (D) 1.11

Answer (B)

Solution .

We know that,

$$\begin{aligned}\text{Respiratory quotient} &= \frac{CO_2 \text{ produced}}{O_2 \text{ consumed}} \\ &= \frac{4.5}{(21 - 16)} \\ &= 0.9\end{aligned}$$

Q 45. Total number of injuries in an opencast coal mine employing 800 persons is 16 in a year . As per DGMS norms, the injury rate per 1000 persons employed is

- (A) 13 (B) 15 (C) 20 (D) 25

Answer (C)

Solution .

$$\begin{aligned}\text{Injury rate per 1000 persons employed} &= \frac{16}{800} \times 1000 \\ &= 20\end{aligned}$$

Q 46. The coefficient of friction between the tub - wheel and haulage track is $1/\sqrt{3}$. for the applicability of direct haulage , minimum inclination (in degrees) of track is

- (A) 60 (B) 55 (C) 35 (D) 30

Answer (D)

Solution .

We have ,

Coefficient of friction = $\tan \theta$

Where,

θ - angle of inclination

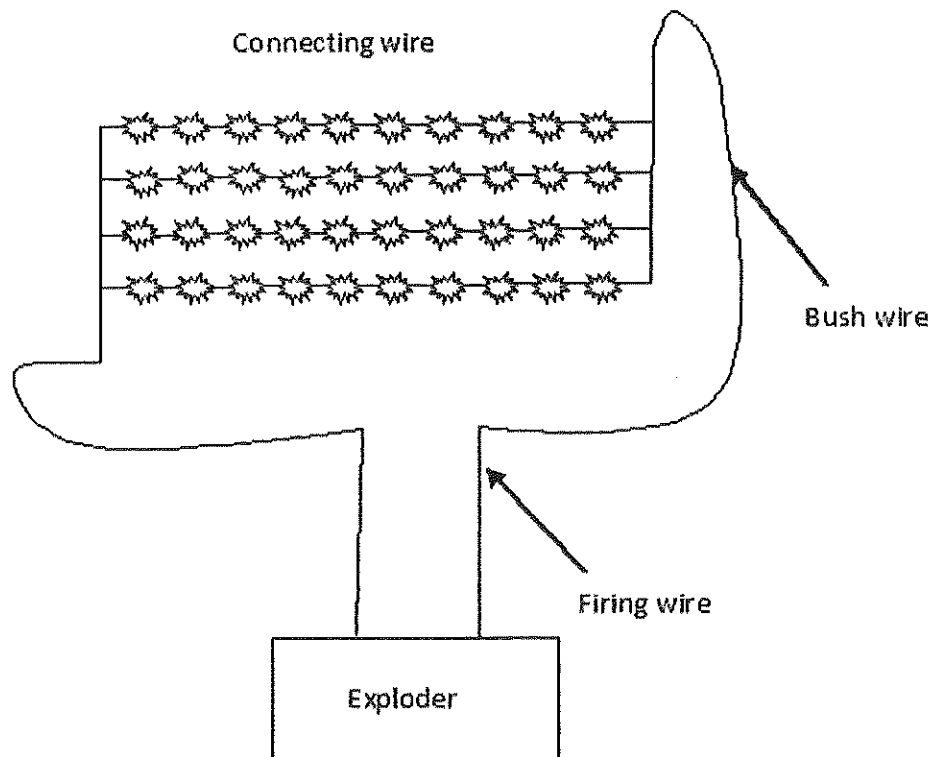
$$\frac{1}{\sqrt{3}} = \tan \theta$$

$$\text{Or } \theta = \tan^{-1} \left(\frac{1}{\sqrt{3}} \right)$$

$$\therefore \theta = 30^\circ$$

Q 47. A surface mine bast pattern shown below has the following details:

accessory	Resistance (in Ohms)	Number of length
Detonator	2 per detonator	40 nos
Connecting wire	0.5 /m	100 m
Bus wire	0.5/m	100 m
Firing line	0.01 /m	200 m



If the exploder supplies 440 V, the current in the blasting circuit in amperes is

- (A) 5.36 (B) 3.51 (C) 4.83 (D) 2.57

Solution .

In the case of the 10 detonators and 25 m connecting wire the resistance will be:-

$$= 2 \times 10 + 25 \times 0.5$$

For 40 detonator and 100 m connecting wire, the resistance will be:-

$$\frac{1}{R} = \frac{1}{32.5} + \frac{1}{32.5} + \frac{1}{32.5} + \frac{1}{32.5}$$

$$\therefore R = 8.125 \text{ ohm}$$

Now, total resistance (R_T) of circuit will be:-

$$R_T = 8.125 + 100 \times 0.5 + 0.01 \times 200$$

$$R_T = 60.125 \text{ ohm}$$

Hence,

The current (I) in the blasting circuit will be:-

$$I = \frac{440}{60.125} = 7.31 \text{ ampere}$$

Q 48. In a surface mine blast, the peak particle velocity (V in mm/s) is estimated from the equation

$V = 120 (\sqrt{SD})^{-1.0}$, where SD is square root sealed distance. If at a distance of 100 m from the blast site the permissible peak particle velocity is 25 mm/s, the maximum charge per delay in kg is

- (A) 404 (B) 414 (C) 434 (D) 464

Answer (C)

Solution .

Given that,

$$V = 120(\sqrt{SD})^{-1.0}$$

$$\text{And } \sqrt{SD} = \frac{D}{\sqrt{Q}}$$

$$\therefore V = 120\left(\frac{D}{\sqrt{Q}}\right)^{-1.0}$$

$$\text{Or } V = 120\left(\frac{\sqrt{Q}}{D}\right)$$

$$25 = 120 \times \frac{\sqrt{Q}}{100}$$

$$\text{Or } \sqrt{Q} = \frac{25 \times 100}{120}$$

$$\therefore Q = 434.03 \text{ kg}$$

Q 49. Daily production measured for a period of 50 days in a coal mine exhibits normal distribution with mean 1200 tpd and standard deviation 100 tpd. The 95 % confidence interval of daily production (standard normal variable Z at 0.025 level of significance is 1.96) in tpd is

- (A) 1200 ± 120.5 (B) 1200 ± 96.0 (C) 1200 ± 39.6 (D) 1200 ± 27.7

Answer (D)

Solution .

We have,

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

$$\text{Or } \bar{X} = \mu \pm z \cdot \frac{\sigma}{\sqrt{n}}$$

$\therefore$ 95 % confidence interval of daily production :-

$$\bar{X} = 1200 \pm 1.96 \times \frac{100}{\sqrt{50}}$$

$$\bar{X} = 1200 \pm 27.71 \text{ ,tpd}$$

- Q 50. In an iron ore deposit alumina is distributed with $\mu = 3\%$ and $\sigma = 0.5\%$, where as silica is distributed with $\mu = 2.5\%$ and $\sigma = 0.8\%$. The combined alumina and silica (as impurities) has μ and σ , in percentage respectively as
- (A) 5.5, 0.94 (B) 5.5, 1.3 (C) 0.5, 0.3 (D) 5.5, 0.52

Answer (A)

Solution .

For alumina - $\mu_1 = 3\%$, $\sigma_1 = 0.5\%$

For silica - $\mu_2 = 2.5\%$, $\sigma_2 = 0.8\%$

Combined mean (μ) = $\mu_1 + \mu_2 = (3 + 2.5)\%$

$$\mu = 5.5\%$$

And combined variance = (combined standard deviation)²

Or combined standard deviation = $\sqrt{\text{combined variance}}$

$$\text{combined standard deviation} = \sqrt{(\text{variation})_{\text{alumina}} + (\text{variation})_{\text{silica}}}$$

$$\text{combined standard deviation} = \sqrt{(\text{standard deviation})_{\text{alumina}}^2 + (\text{standard deviation})_{\text{silica}}^2}$$

$$\therefore \text{combined standard deviation} = \sqrt{0.5^2 + 0.8^2}$$

$$\therefore \text{combined standard deviation} = 0.94\%$$

- Q 51. The inverse of the following matrix is:

$$\begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(A) $\begin{bmatrix} 16 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} 0.25 & 0 & 0 \\ 0 & 0.50 & 0 \\ 0 & 0 & 1.00 \end{bmatrix}$

(C) $\begin{bmatrix} 2 & 0 & 0 \\ 0 & \sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(D) $\begin{bmatrix} 16 & 0 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

Answer (B)

Solution .

$$A = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Now, } |A| = \begin{vmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$\therefore |A| = 8$$

$$\text{Here, } A_{11} = (-1)^{1+1} \begin{vmatrix} 2 & 0 \\ 0 & 1 \end{vmatrix} = 2$$

$$A_{12} = (-1)^{1+2} \begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix} = 0$$

$$A_{13} = (-1)^{1+3} \begin{vmatrix} 0 & 2 \\ 0 & 0 \end{vmatrix} = 0$$

$$A_{21} = (-1)^{2+1} \begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix} = 0$$

$$A_{22} = (-1)^{2+2} \begin{vmatrix} 4 & 0 \\ 0 & 1 \end{vmatrix} = 4$$

$$A_{23} = (-1)^{2+3} \begin{vmatrix} 4 & 0 \\ 0 & 0 \end{vmatrix} = 0$$

$$A_{31} = (-1)^{3+1} \begin{vmatrix} 0 & 0 \\ 2 & 0 \end{vmatrix} = 0$$

$$A_{32} = (-1)^{3+2} \begin{vmatrix} 4 & 0 \\ 0 & 0 \end{vmatrix} = 0$$

$$A_{33} = (-1)^{3+3} \begin{vmatrix} 4 & 0 \\ 0 & 2 \end{vmatrix} = 8$$

$$\therefore \text{adj } A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 8 \end{bmatrix}$$

Hence,

$$A^{-1} = \frac{\text{adj } A}{|A|} = \frac{\begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 8 \end{bmatrix}}{8}$$

$$A^{-1} = \begin{bmatrix} 0.25 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 1.00 \end{bmatrix}$$

Q 52. The solution of the following system of linear equations is:

$$x + 4y + 3z = 0$$

$$3x + 5y + 2z = 0$$

$$8x + 10y + 12z = 0$$

- (A) (0,0,0) (B) (1,-1,1) (C) (2, -1, -2) (D) (-3,0,1)

Answer (A)

Solution .

Now,

$$\Delta = \begin{vmatrix} 1 & 4 & 3 \\ 3 & 5 & 2 \\ 8 & 10 & 12 \end{vmatrix}$$

$$\Delta = 1 \begin{vmatrix} 5 & 2 \\ 10 & 12 \end{vmatrix} - 4 \begin{vmatrix} 3 & 2 \\ 8 & 12 \end{vmatrix} + 3 \begin{vmatrix} 3 & 5 \\ 8 & 10 \end{vmatrix}$$

$$\Delta = 40 - 4 \times 20 - 3 \times 10$$

$$\Delta = -70$$

Here, $\Delta \neq 0$

Hence, given linear equation has trivial solution, So

$$x = 0, y = 0, z = 0$$

Q 53. The volume of a cone is given by

$$V = \frac{\pi}{3} l^3 \sin^2 \theta \cos \theta$$

Where, l is the slant height and θ is the semi-vertical angle. The angle (θ), for which the volume of cone becomes maximum is

- (A) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (B) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$
(C) $\cos^{-1}(\sqrt{2})$ (D) $\sin^{-1}(\sqrt{2})$

Answer (B)

Solution .

$$V = \frac{\pi}{3} l^3 \cdot \sin^2 \theta \cdot \cos \theta \dots\dots\dots(1)$$

Differentiating equa. (1) both sides with respect to θ ,

$$\frac{dV}{d\theta} = \frac{\pi}{3} l^3 [2 \sin \theta \cos \theta \cos \theta - \sin^2 \theta \sin \theta]$$

$$\frac{dV}{d\theta} = \frac{\pi}{3} l^3 [2 \sin \theta \cos^2 \theta - \sin^2 \theta \sin \theta]$$

The volume of cone becomes maximum ,when

$$\frac{dV}{d\theta} = 0$$

Therefore,

$$\frac{\pi}{3} l^3 [2 \sin \theta \cos^2 \theta - \sin^2 \theta \sin \theta] = 0$$

$$\therefore [2 \sin \theta \cos^2 \theta - \sin^2 \theta \sin \theta] = 0$$

$$\text{Or } 2\cos^2 \theta - \sin^2 \theta = 0$$

$$\text{Or } 2\cos^2 \theta - 1 + \cos^2 \theta = 0 \quad [\text{since } \sin^2 \theta + \cos^2 \theta = 1]$$

$$\text{Or } 3\cos^2 \theta = 1$$

$$\therefore \cos \theta = \frac{1}{\sqrt{3}}$$

$$\therefore \theta = \cos^{-1} \left(\frac{1}{\sqrt{3}} \right)$$

Q 54. The direction of gradient vector at a point (1,1,2) on a surface $S(x, y, z) = x^2 + y^2 - z$

(A) $\frac{1}{3}(2\hat{i} - 2\hat{j} + \hat{k})$

(B) $\frac{1}{3}(-2\hat{i} + 2\hat{j} + \hat{k})$

(C) $\frac{1}{3}(2\hat{i} - 2\hat{j} + \hat{k})$

(D) $\frac{1}{3}(2\hat{i} + 2\hat{j} - \hat{k})$

Answer (D)

Solution .

$$\nabla.S = \left(I \frac{\partial}{\partial x} + J \frac{\partial}{\partial y} + K \frac{\partial}{\partial z} \right) \cdot (x^2 + y^2 - z)$$

$$\nabla.S = I \frac{\partial}{\partial x} (x^2 + y^2 - z) + J \frac{\partial}{\partial y} (x^2 + y^2 - z) + K \frac{\partial}{\partial z} (x^2 + y^2 - z)$$

$$\nabla.S = 2x \mathbf{i} + 2y \mathbf{j} - \mathbf{k}$$

At point (1,1,2)

$$\nabla.S = 2\mathbf{i} + 2\mathbf{j} - \mathbf{k}$$

$$\text{Now, direction of the gradient vector} = \frac{2\mathbf{i} + 2\mathbf{j} - \mathbf{k}}{\sqrt{4+4+1}} = \frac{1}{3}(2\mathbf{i} + 2\mathbf{j} - \mathbf{k})$$

Q 55. The solution of the differential equation

$$\frac{d^2y}{dx^2} + 3 \frac{dy}{dx} - 4y = 0, \quad \text{is}$$

$$(A) \quad y = c_1 e^{4x}$$

$$(B) \quad y = c_1 e^{2x}$$

$$(C) \quad y = c_1 e^x + c_2 e^{-4x}$$

$$(D) \quad y = c_1 e^{-x} + c_2 e^{4x}$$

Answer (C)

Solution .

Given differential equation is-

$$\frac{d^2y}{dx^2} + 3 \frac{dy}{dx} - 4y = 0,$$

In symbolic form-

$$D^2y + 3Dy - 4y = 0$$

$$(D^2 + 3D - 4)y = 0$$

It's A.E. will be

$$D^2 + 3D - 4 = 0$$

$$D^2 + 4D - D - 4 = 0$$

$$\therefore (D+4).(D-1) = 0$$

$$\therefore D = 1, -4$$

Hence, solution will be-

$$y = c_1 e^x + c_2 e^{-4x}$$

Q 56. A force vector $= (2\hat{i} + 3\hat{j} - \hat{k})$ in N is acting on a point, whose position vector $r = (\hat{i} - \hat{j} - 6\hat{k})$ in m. The magnitude of the torque about the origin in Nm is

- (A) 20.85 (B) 21.42 (C) 21.97 (D) 22.27

Answer (C)

Solution .

$$\vec{F} = 2\hat{i} + 3\hat{j} - \hat{k}$$

$$\vec{r} = \hat{i} - \hat{j} + 6\hat{k}$$

$$\therefore \text{Torque} = \vec{r} \times \vec{F}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 6 \\ 2 & 3 & -1 \end{vmatrix}$$

$$= \hat{i} \begin{vmatrix} -1 & 6 \\ 3 & -1 \end{vmatrix} - \hat{j} \begin{vmatrix} 1 & 6 \\ 2 & -1 \end{vmatrix} + \hat{k} \begin{vmatrix} 1 & -1 \\ 2 & 3 \end{vmatrix}$$

$$\text{Torque} = -17\hat{i} + 13\hat{j} + 5\hat{k}$$

$$\begin{aligned} \text{Now, Magnitude of the torque} &= \sqrt{(-17)^2 + (13)^2 + (5)^2} \\ &= 21.97 \text{ Nm} \end{aligned}$$

Q 57. If H is the maximum height attained by a projectile, the maximum horizontal range when fired at 45° inclination from the ground level is

- (A) 4.0 H (B) 3.6 H (C) 3.2 H (D) 2.7 H

Answer (A)

Solution .

We have,

$$R_{\max} = \frac{u^2}{g} \dots\dots\dots(1)$$

$$\text{And } H_{\max} = \frac{u^2 (\sin \theta)^2}{2g} \dots\dots\dots(2)$$

From equation (1) & (2)

$$\frac{R}{H} = \frac{2}{(\sin \theta)^2}$$

$$\frac{R}{H} = \frac{2}{(\sin 45^\circ)^2}$$

$$\therefore R = 4H$$

Q 58. The local mean time at longitude $75^\circ 30'$ is 8 hr 45 min. The corresponding standard time with reference to $82^\circ 30'$ meridian is

- (A) 8 hr 13 min (B) 9 hr 13 min (C) 9 hr 17 min (D) 10 hr 17 min

Answer (B)

Solution .

For every $15^\circ \rightarrow 1$ hour

$$\therefore 1^\circ \rightarrow \frac{1}{15} \times 60 = 4 \text{ min.}$$

Now,

$$\begin{aligned} \text{Change in meridian} &= 82^\circ 30' - 75^\circ 30' \\ &= 7^\circ \end{aligned}$$

$$\therefore \text{Change in time} = 7 \times 4 = 28 \text{ min.}$$

Therefore,

Standard time with reference to $82^\circ 30'$ Meridian :-

$$= 8 \text{ hr } 45 \text{ min} + 0 \text{ hr } 28 \text{ min.}$$

$$= 8 \text{ hr } 73 \text{ min.}$$

$$= 9 \text{ hr } 13 \text{ min.}$$

Q 59. Block economic values in 2D block model are shown below. Then based on the assumption of 1:1 slope angle. The blocks (identified by row and column numbers) that constitute the ultimate pit are

-1	-1	1	-1
-1	1	3	-1
-1	-1	-1	-1

(A) (1,2),(1,2),(1,3),(2,2) (B) (1,2),(1,3),(1,4),(2,3)

(C) (1,3),(2,4) (D) (1,3),(1,4),(2,4)

Answer (B)

Solution .

The following steps are used-

- (1) The cone is floated from left to right along the top row of the blocks in the section, if there is a positive block it is removed.

	1	2	3	4	
1	-1	-1		-1	+1
2	-1	1	3	-1	
3	-1	-1	-1	-1	

- (2) After traversing the first row the apex of the cone is moved to the second row, starting floats from left to right stopping when it encounters the first positive blocks. If the sum of all blocks falling within the cone is positive (or zero), these blocks are removed . if the sum is negative the blocks are left and thee cone floats to the next positive blocks on this row. The summing and mining or leaving process is repeated.

	1	2	3	4	
1				-1	-1-1+1=-1
2	-1		3	-1	
3	-1	-1	-1	-1	

	1	2	3	4	
1	-1				$-1-1+3 = +1$
2	-1	1		-1	
3	-1	-1	-1	-1	

(3) This floating cone process moving from left to right and top to bottom of the section continues until no more blocks can be removed.

(4) The profitability of the section is found by summing the value of the blocks removed.

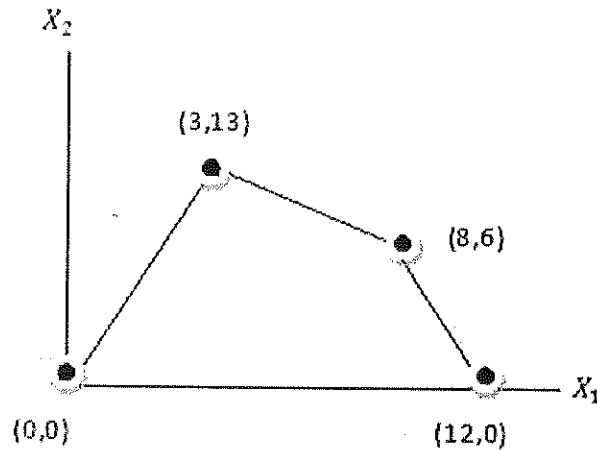
∴ The optimum pit value of section = $+1 + 1 = +2$

ultimate pit limit
↓

	1	2	3	4
1	-1	-1	1	-1
2	-1	1	3	-1
3	-1	-1	-1	-1

Therefore, Blocks (1, 2), (1, 3), (1, 4) and (2, 3) constitute a ultimate pit limit.

Q 60. The feasible region of an LP problem is shown as given below. The maximum value of the objective function $Z = 1600x_1 + 1200x_2$ is



- (A) 20400 (B) 20000 (C) 19200 (D) 16800

Answer (A)

Solution .

We have,

$$Z = 1600X_1 + 1200X_2$$

Maximum value of objective function will be at point (3,13).

$$Z_{max} = 1600 \times 3 + 1200 \times 13$$

$$Z_{max} = 20400$$

- Q 61.** A conveyor of rated power 100 kW hauls up –dip at 30 kg/s along an inclination of 15° and distance 300 m. Heat added by the conveyor to the air in kW is

- (A) 56.4 (B) 65.9 (C) 77.2 (D) 82.3

Answer (C)

Solution .

$$\text{Component of weight along incline plane} = mg \sin \theta$$

$$= 30 \times 9.81 \times \sin 15^\circ$$

$$= 76.17 \text{ N}$$

and Work done = force $\times$ distance

$$= 76.17 \times 300$$

$$= 22851.13 \text{ joules} = 22851.13 \text{ watt} = 22.85 \text{ kW}$$

Hence, Heat added by the conveyor to the air = $100 - 22.85$

$$= 77.2 \text{ kW}$$

Q 62. A cage of floor area 5 m^2 suspended in a shaft has a drag coefficient of 2.5. if the velocity of air in the shaft is 6 m/s , the drag force (N) experienced by the cage is

(A) 120

(B) 170

(C) 200

(D) 270

Answer (D)

Solution .

We have,

$$\text{Drag force} = \frac{1}{2} \times \rho \times v^2 \times C_D \times A \times g, \text{ N}$$

$$\text{Where, } \rho = \text{density} = 1.2 \text{ kg/m}^3$$

$$V = \text{velocity} = 6 \text{ m/s}$$

$$C_D = \text{Drag coefficient} = 2.5$$

$$A = \text{cross sectional area} = 5 \text{ m}^2$$

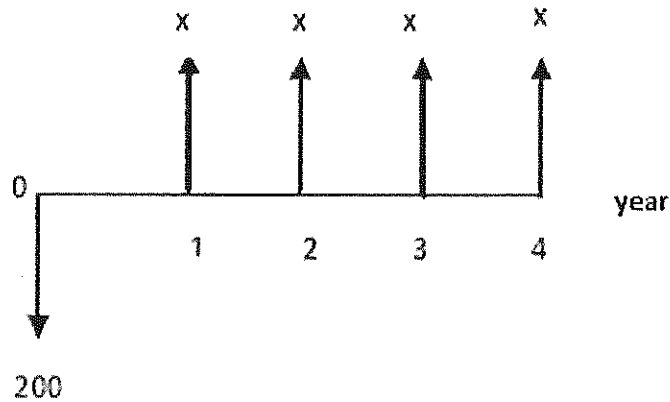
$$g = \text{acceleration due to gravity} = 9.81 \text{ m/s}^2$$

Hence,

$$\text{Drag force} = \frac{1}{2} \times 1.2 \times 6^2 \times 2.5 \times 5 \times 9.81$$

$$\text{Drag force} = 270 \text{ N}$$

Q 63. A cash flow diagram is shown below. Based on NPV, at 10% rate of interest, the minimum annuity 'x' at which the instrument become viable



(A) 63

(B) 54

(C) 42

(D) 35

Answer (A)

Solution .

We know,

$$\text{NPV} = - \text{investment} + \text{annual payment (X)} \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right]$$

Investment would be acceptable ,when

$$\text{NPV} \geq 0$$

$$-200 + (X) \left[\frac{(1+0.1)^4 - 1}{0.1(1+0.1)^4} \right] \geq 0$$

$$\therefore -200 + (X) \cdot 3.17 \geq 0$$

$$\therefore (X) \geq \frac{200}{3.17}$$

$$\therefore (X) = 63$$

Q 64. A system of two identical mine pumps connected in series has reliability 0.49. if the pump were to be connected in parallel, the system reliability would be

(A) 0.21

(B) 0.6

(C) 0.91

(D) 0.95

Answer (C)

Solution .

If R_1 and R_2 are the reliabilities of two components respectively and combined in series then

System reliability (R) is given by:-

$$R = R_1 \times R_2$$

$$\therefore R_1 = R_2$$

$$\therefore 0.49 = R_1 \times R_1$$

$$\therefore R_1 = 0.7 \text{ \& } R_2 = 0.7$$

If R_1 and R_2 are the reliabilities of two components respectively and combined in parallel then System reliability (R') is given by:-

$$\therefore R' = (1 - R_1) \times (1 - R_2)$$

$$R' = 1 - (1 - 0.7) \times (1 - 0.7)$$

$$\therefore R' = 0.91$$

Q 65. An SDL working at different faces gives the following performance:

Operating faces	production per blast (tonne)	Muck clearance time (hrs)
Development face	16	1.25
Splitting face	17	1.20
Slicing faces	18	1.30
Heightening faces	20	1.50

In 5 hours operation maximum output is obtained from the

- (A) Heightening faces (B) Slicing faces
(C) Development faces (D) Splitting face

Answer (B)

Solution .

Operating face	Output in 5 hours operation (tonnes)
Development face	$\frac{5 \times 16}{1.25} = 64$
Splitting face	$\frac{5 \times 17}{1.20} = 70.84$
Slicing face	$\frac{5 \times 18}{1.30} = 69.23$
Heighting face	$\frac{5 \times 20}{1.50} = 66.67$

∴ maximum output from from Splitting face.

Q 66. In a coal handling plant wagons of 8 m length are loaded at rake travel speed of 0.48 km/hr. The chute loading rate is 6000 tonne/hr. As the rake moves continuously, the chute stops for a total of 24 s in between two wagons. The quantity of coal in tonne loaded in each wagon is

- (A) 52 (B) 60 (C) 76 (D) 94

Answer (B)

Solution .

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

$$\therefore \frac{0.48 \times 1000}{3600} = \frac{8}{\text{time}}$$

$$\therefore \text{time} = 60 \text{ sec.}$$

Now, time taken to fill a wagon = 60 – 24 = 36 sec.

But given chute loading rate = 6000 tonne/hr

Hence,

Quantity of coal loaded in each wagon = loading rate × time taken to fill a wagon

$$= 6000 \frac{\text{tonne}}{\text{h}} \times 36 \text{ sec}$$

$$= 6000 \frac{\text{tonne}}{3600 \text{ sec}} \times 36 \text{ sec}$$

$$= 60 \text{ tonnes}$$

Q 67. Match the following

Failure criteria

Relationship

P. Drucker-Prager 1. $\sigma_1 = \sigma_3 + \sqrt{m \sigma_3 + s^2}$

Q. Hoek-Brown 2. $\tau = c + \sigma_n \tan \phi$

R. Mohr-Coulomb 3. $\sqrt{\frac{2}{3} \left[\left(\frac{\sigma_1 - \sigma_2}{2} \right)^2 + \left(\frac{\sigma_2 - \sigma_3}{2} \right)^2 + \left(\frac{\sigma_3 - \sigma_1}{2} \right)^2 \right]} =$

$$A (\sigma_1 + \sigma_2 + \sigma_3) + B$$

(A) P-1, Q-3, R-2

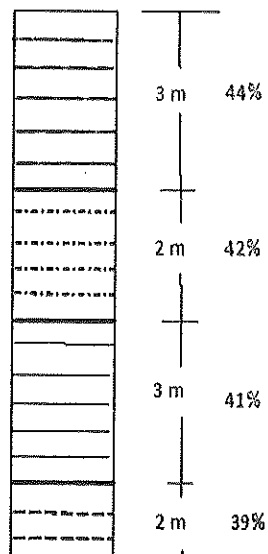
(B) P-3, Q-1, R-2

(C) P-3, Q-2, R-1

(D) P-1, Q-2, R-3

Answer (B)

Q 68. An assay value of alumina in a borehole from a bauxite deposit is shown in the figure. If the cut off grade is 40 %, the composite value of ore in the borehole in percent is



(A) 31.6

(B) 33.9

(C) 41.7

(D) 42.2

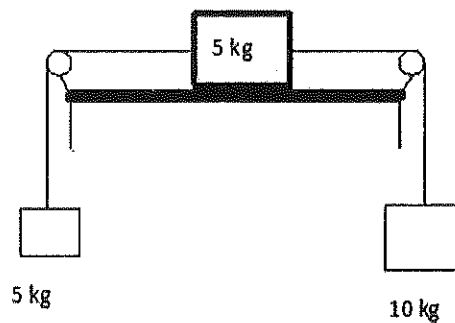
Answer (C)

Solution .

We have,

$$\begin{aligned}\text{Composite value of ore grade} &= \frac{L_1 g_1 + L_2 g_2 + L_3 g_3 + L_4 g_4}{L_1 + L_2 + L_3 + L_4} \\ &= \frac{3 \times 44 + 2 \times 42 + 3 \times 41 + 2 \times 39}{3 + 2 + 3 + 2} \\ &= 41.7 \%\end{aligned}$$

Common data for questions 69,70 and 71: Two blocks of mass 5 kg and 10 kg are connected with cords and frictionless pulley as shown. Friction coefficient between the 5 kg and table is 0.2



Q 69. The acceleration of the system when the blocks are released from rest ('g' is acceleration due to gravity)

(A) 5g

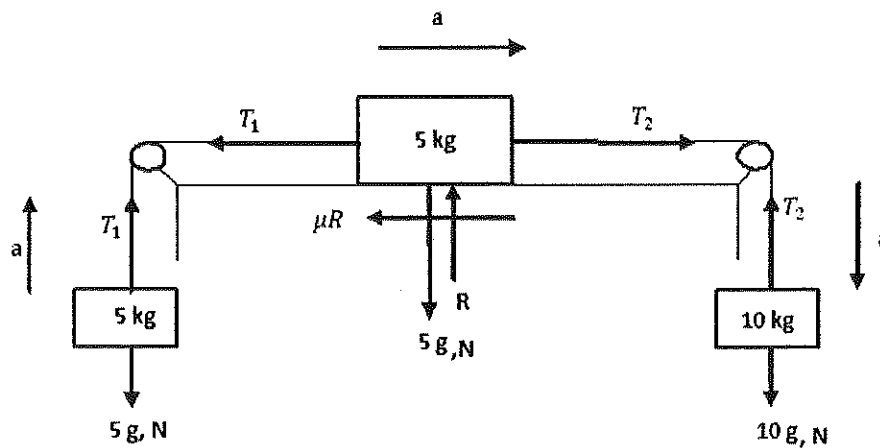
(B) 2g

(C) g/5

(D) g/10

Answer (C)

Solution .



From left hand side mass, applying $F = ma$, then

$$T_1 - 5g = 5a$$

$$\text{Or } T_1 = 5a + 5g \dots\dots\dots(i)$$

From right hand side mass, applying $F = ma$, then

$$10g - T_2 = 10a$$

$$\text{Or } T_2 = 10g - 10a \dots\dots\dots(ii)$$

From middle mass applying $F = ma$, then

$$T_2 - T_1 - \mu R = 5a$$

But when $\sum V = 0$ then, $R = 5g$

$$\therefore T_2 - T_1 - 5g \times 0.2 = 5a$$

$$\therefore T_2 - T_1 - g = 5a \dots\dots\dots(iii)$$

Putting the value of T_1 & T_2 in equation (iii)

$$10g - 10a - 5a - 5g - g = 5a$$

$$\therefore a = \frac{g}{5}$$

Q 70. Tension (N) in the cord connected to the 10 kg block is

- (A) $8g$ (B) $6g$ (C) $4g$ (D) $2g$

Answer (A)

Solution .

From equation (2)

$$T_2 = 10 g - 10 \times \frac{g}{5}$$

$$T_2 = 8 g, N$$

Q 71. Tension (N) in the cord connected to the 5 kg block is

(A) 8g (B) 6g (C) 4g (D) 2g

Answer (B)

Solution .

From equation (1)

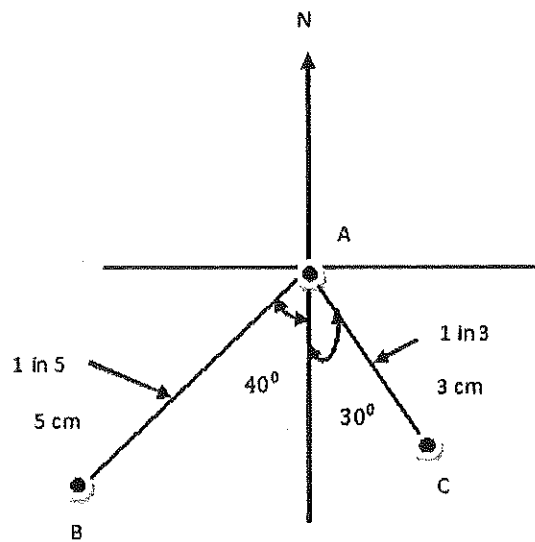
$$T_1 = 5 a + 5 g = 5 \times \frac{g}{5} + 5g$$

$$T_1 = 6 g, N$$

Common data for questions 72 and 73: Three boreholes intersect a coal seam at a points A, B and C as shown (figure is drawn to scale)

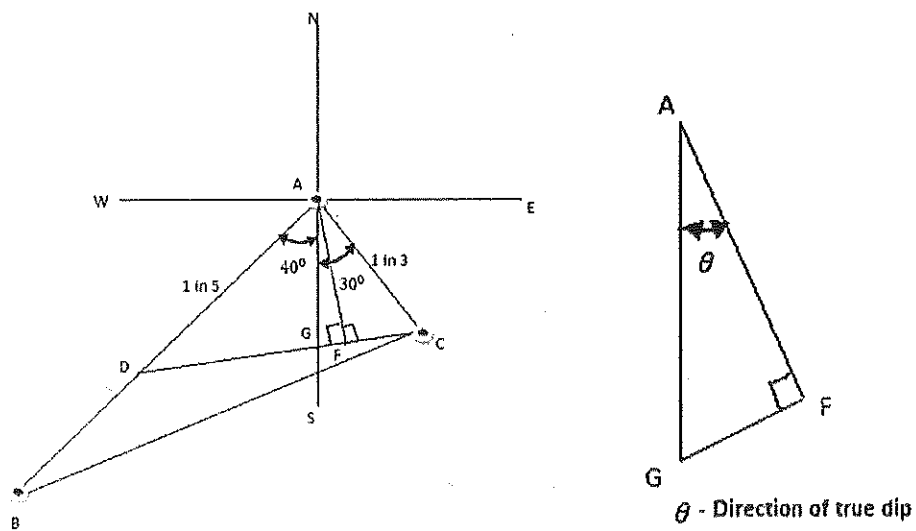
The survey details are given below

Line	Bearing	Gradient
AB	$S40^{\circ}W$	1 in 5
AC	$S30^{\circ}E$	1 in 3



- Q 72. The direction of the true dip of the seam is
- (A) $S15^{\circ}W$ (B) $S25^{\circ}W$ (C) $S15^{\circ}E$ (D) $S30^{\circ}E$
- Answer (C)

Solution .



We have,

In $\triangle ACD$

$$\tan \frac{D-C}{2} = \frac{c-d}{c+d} \tan \frac{D+C}{2}$$

Now,

$$A + C + D = 180^\circ$$

$$\text{But } \angle A = 70^\circ$$

$$\therefore C + D = 110^\circ \dots\dots\dots(1)$$

Now,

$$\tan \frac{D-C}{2} = \frac{3-5}{3+5} \tan \frac{110^\circ}{2}$$

On solving, it is found that

$$D - C = -39^\circ 35' 91'' \dots\dots\dots(2)$$

Solving (1) & (2)

$$\angle D = 35^\circ 12' 05''$$

$$\angle C = 74^\circ 47' 55.95''$$

In $\triangle AGC$

$$A + C + G = 180^\circ$$

$$\angle G = 180^\circ - 30^\circ - 74^\circ 47' 55.95''$$

$$\angle G = 75^\circ 12' 4.44''$$

In $\triangle AGF$

$$A + G + F = 180^\circ$$

$$\angle A = 180^\circ - 90^\circ - 75^\circ 12' 4.44''$$

$$\angle A = 14^\circ 47' 55.56''$$

Hence,

Direction of true dip of seam is $S14^\circ 47' 55.56''E$ or $S15^\circ E$

Q 73. The gradient of the seam

- (A) 1 in 2.7 (B) 1 in 3.7 (C) 1 in 4.7 (D) 1 in 5.7

Answer (A)

Solution .

We know that,

$$\text{Apparent dip} = \text{full dip} \times \cos \alpha$$

α - angle between full dip and apparent dip

$$\frac{1}{5} = \text{full dip} \times \cos(40^\circ + 14^\circ 47' 55.56'')$$

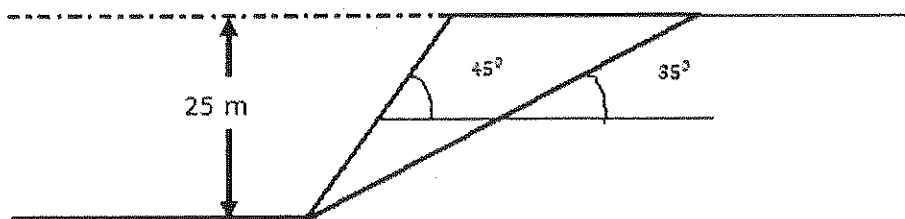
$$\therefore \text{full dip or gradient of the seam} = \frac{1}{2.7}$$

Or

$$\frac{1}{3} = \text{full dip} \times \cos(30^\circ - 14^\circ 47' 55.56'')$$

$$\therefore \text{full dip or gradient of the seam} = \frac{1}{2.7}$$

Statement for linked answer questions 74 and 75: An opencast mine bench has a potential failure plane as indicated below. The unit weight, cohesion, and angle of internal friction of the rock mass are 24 kN/m^3 , 0.02 MPa and 30° respectively.

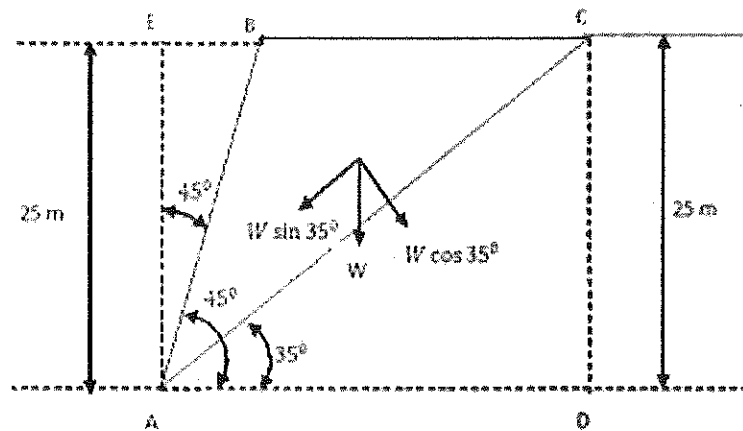


Q 74. The driving force for failure, on the potential failure plane is

- (A) 187 N (B) 1.87 kN (C) 18.7 kN (D) 1.87 MN

Answer (D)

Solution .



$$\tan 35^\circ = \frac{25}{AD}$$

$$\text{Or } AD = \frac{25}{\tan 35^\circ} = 35.70 \text{ m}$$

$$\text{Now, area of triangle ADC} = \frac{1}{2} \times 25 \times 35.70 = 446.25 \text{ m}^2$$

$$\text{Again } \tan 45^\circ = \frac{EB}{25}$$

$$EB = 25 \text{ m}$$

$$\text{Now, area of triangle ABE} = \frac{1}{2} \times 25 \times 25 = 312.5 \text{ m}^2$$

$$\text{And area of rectangle ACDE} = 35.70 \times 25 = 892.5 \text{ m}^2$$

$$\begin{aligned} \therefore \text{area of triangle ABC} &= 892.5 - 446.25 - 312.5 \\ &= 133.75 \text{ m}^2 \end{aligned}$$

$$\text{Given that, unit weight} = 24.5 \text{ kN/m}^3$$

$$\text{So, weight of block (W)} = 24.5 \times 133.75 \times 1$$

$$W = 3276.875 \text{ kN}$$

Now, driving force for failure is-

$$\begin{aligned} W \sin \beta &= 3276.875 \times \sin 35^\circ \\ &= 1.87 \text{ MN} \end{aligned}$$

Q 75. The 'factor of safety' of slope under given condition is

- (A) 0.7 (B) 0.9 (C) 1.1 (D) 1.3

Answer (D)

Solution .

$$\text{Factor of safety of slope (FOS)} = \frac{C.A + W \cos \alpha \tan \phi}{W \sin \alpha}$$

$$\text{Here, } \alpha = 35^\circ$$

$$\phi = 30^\circ$$

$$\text{Cohesion} = C = 0.02 \text{ MPa}$$

$$\text{Shear area (A)} = \left(\frac{25}{\sin 35^\circ} \right) \times 1, \text{ m}^2$$

$$W = 3276.88 \text{ kN} = 3.28 \text{ MN}$$

$$\therefore \text{FOS} = \frac{0.02 \times \left(\frac{25}{\sin 35^\circ} \right) \times 1 + 3.28 \times \cos 35^\circ \times \tan 30^\circ}{3.28 \times \sin 35^\circ}$$

$$\therefore \text{FOS} = 1.3$$

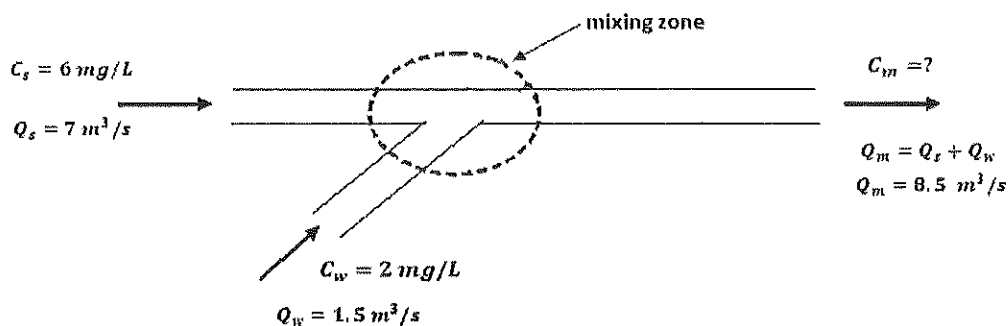
Statement for linked answer questions 76 and 77: Mine water flowing at $1.5 \text{ m}^3/\text{s}$ with 2 mg/l dissolved oxygen, joins river water flowing at $7 \text{ m}^3/\text{s}$ containing 6 mg/l dissolved oxygen

Q 76. The dissolved oxygen concentration of mixture in mg/l is

- (A) 5.3 (B) 4.8 (C) 4.2 (D) 3.9

Answer (A)

Solution .



$$C_m = \frac{C_s Q_s + C_w Q_w}{Q_m} = \frac{6 \times 7 + 2 \times 1.5}{8.5}$$

$$C_m = 5.3 \text{ mg/L}$$

Q 77. The saturated value of the dissolved oxygen in the mixture is given to be 9.3 mg/l. On this basis the initial oxygen deficit of the mixture in mg/l is

- (A) 2.4 (B) 4.0 (C) 6.8 (D) 14.6

Answer (B)

Solution .

Initial oxygen deficit of mixture = saturated value of dissolved oxygen in mixture—dissolved oxygen concentration of mixture

$$= 9.3 - 5.3$$

$$= 4 \text{ mg/l}$$

Statement for linked answer questions 78 and 79:

Unit cost matrix of a transportation problem is given below in certain monetary unit.

		Destination			
		1	2	3	
Source	1	2	2	8	15
	2	1	5	7	40
	3	6	4	3	20
		10	25	40	Demand

Q 78. The total cost of transportation based on the initial feasible solution obtained by the

North- West corner rule is

- (A) 250 (B) 290 (C) 330 (D) 360

Answer (C)

Solution .

2 (10)	2 (5)	8	15/5/0
1	5 (20)	7 (20)	40/20/0 supply
6	4	3 (20)	20/0
10/0	25/20/0	40/20/0	Demand

$$\begin{aligned} \text{Total cost of transportation} &= 2 \times 10 + 2 \times 5 + 5 \times 20 + 7 \times 20 + 3 \times 20 \\ &= 330 \end{aligned}$$

Q 79. The optimal solution for the transportation problem has allocation as shown below;

		Destination			
		1	2	3	
Source	1		15		15
	2	10	10	20	40 Supply
	3			20	20
		10	25	40	Demand

When compared to initial basic feasible solution from the above, the optimal allocation

results in savings of

- (A) 10 (B) 20 (C) 30 (D) 40

Answer (D)

Solution .

	15		15	
10	10	20	40	supply
		20	20	
10	25	40		Demand

$$\begin{aligned}\text{Here , Optimal cost of transportation} &= 15 \times 2 + 10 \times 1 + 10 \times 5 + 20 \times 7 + 20 \times 3 \\ &= 290\end{aligned}$$

$$\text{Hence , cost of saving} = 330 - 290 = 40$$

Statement for linked answer questions 80 and 81: In a mine ventilation system, the resistance of two splits A and B are $0.5 \text{ N s}^2 \text{ m}^{-8}$ and $2.0 \text{ N s}^2 \text{ m}^{-8}$ respectively.

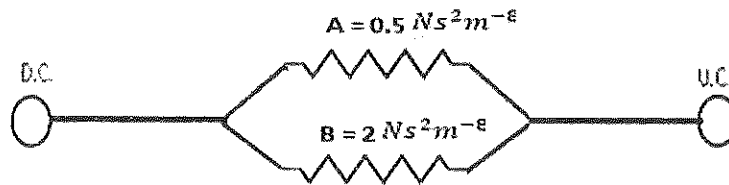
Combined resistance of two shafts and trunk roadways is $0.7 \text{ N s}^2 \text{ m}^{-8}$. A quantity of $20 \text{ m}^3/\text{s}$ of air passé through split A

Q 80. The total air quantity passing the mine in m^3/s

- (A) 30 (B) 27 (C) 25 (D) 7

Answer (A)

Solution .



$$\frac{1}{\sqrt{R_{AB}}} = \frac{1}{\sqrt{R_A}} + \frac{1}{\sqrt{R_B}}$$

$$\frac{1}{\sqrt{R_{AB}}} = \frac{1}{\sqrt{0.5}} + \frac{1}{\sqrt{2}}$$

$$\therefore R_{AB} = 0.223 \text{ N s}^2 \text{ m}^{-8}$$

$$\text{Now, total resistance} = 0.7 + 0.223 = 0.923 \text{ N s}^2 \text{ m}^{-8}$$

$$\text{And } P_A = P_B$$

$$R_A(Q_A)^2 = R_B(Q_B)^2$$

$$0.5 \times (20)^2 = 2 \times (Q_B)^2$$

$$\therefore Q_B = 10 \text{ m}^3/\text{sec}$$

$$\text{Hence, total quantity} = 20 + 10 = 30 \text{ m}^3/\text{sec}$$

Q 81. The total air power of the ventilation system in kW is

- (A) 82.9 (B) 48.9 (C) 24.9 (D) 27.9

Answer (C)

Solution .

We have,

$$\text{Air Power} = PQ \times 10^{-3}, \text{ KW}$$

$$\text{Or Air Power} = RQ^3 \times 10^{-3}, \text{ KW}$$

$$= 0.923 \times (30)^3 \times 10^{-3}$$

$$\text{Air Power} = 24.9 \text{ KW}$$

Statement for linked answer questions 82 and 83: A loco of mass 10000 kg has a coefficient of adhesion to the tracks as 0.25. the offers a rolling resistance equal to 10 % of its weight.

Q 82. The draw- bar- pull generated by the loco on a level ground in KN is

- (A) 11.3 (B) 14.7 (C) 15.8 (D) 17.2

Answer (B)

Solution .

$$\text{Mass of loco} = 10000 \text{ kg} = 10000 \times 9.81 = 98100 \text{ N}$$

$$\text{Tractive effort} = \text{coefficient of adhesion} \times \text{mass}$$

$$= 0.25 \times 98100$$

$$= 24524 \text{ N}$$

$$\text{And Running Resistance} = \frac{10}{100} \times 98100 = 9810 \text{ N}$$

Now, draw bar pull generated by the loco on a level gradient = Tractive effort — Running Resistance

$$= 24525 - 9810$$

$$= 14715 \text{ N}$$

$$= 14.7 \text{ kN}$$

Q 83. The draw-bar-pull generated by the loco when the upward gradient of the track is 5° in kN is

- (A) 6.16 (B) 7.9 (C) 9.5 (D) 1.5

Answer (A)

Solution .

Draw bar pull generated by loco, when the upward gradient of track is 5° = Tractive effort — running resistance — component of force on inclined track

$$= 24525 - 9810 - 1000 \times 9.81 \times \sin 5^\circ$$

$$= 6.16 \text{ kN}$$